Stanley Supplementary Exercise 83(b)

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Original statement

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83. Let $\omega_y : \Lambda(x,y) \to \Lambda(x) \otimes \Lambda(y)$ be the algebra homomorphism defined by

$$\omega_y p_n(x,y) = p_n(x) + (-1)^{n-1} p_n(y). \tag{4}$$

Equivalently, ω_y is the automorphism ω acting on the y-variables only. Write $\omega_y f(x,y) = f(x/y)$. In particular, $s_\lambda(x/y)$ is called a *super Schur function*. Let

$$\Sigma = \mathrm{im}(\omega_y) = \{f(x/y) : f \in \Lambda\},$$

a subalgebra of $\Lambda(x) \otimes \Lambda(y)$.

(b) [3–] Let $g(x,y) \in \Lambda(x) \otimes \Lambda(y)$, and let t be an indeterminate. Show that $g \in \Sigma$ if and only if

$$g(x,y)|_{x_1=t,\, y_1=-t} = g(x,y)|_{x_1=y_1=0}\,. \tag{6}$$

Original solution

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(b) It is clear from the definition (4) that

$$p_n(x/y)_{x_1=t,\,y_1=-t} = p_n(x/y)_{x_1=y_1=0}.$$

Since ω_y and the substitutions $x_1 = t$, $y_1 = -t$ and $x_1 = y_1 = 0$ are homomorphisms (into suitable rings), it follows that $g(x, y)$ satisfies (6) whenever $g \in \mathrm{im}(\omega_y)$.

Conversely, suppose that $g(x, y) \in \Lambda(x) \otimes \Lambda(y)$ and that $g(x, y)$ satisfies (6). Thus by hypothesis

$$g((t, x), (-t, y)) = g(x, y), \tag{25}$$

where $(t, x) = (t, x_1, x_2, \ldots)$ and similarly for (t, y). Define

$$u_i = \frac{1}{2}\Big(p_i(x) + (-1)^i p_i(y)\Big)$$
$$v_i = \frac{1}{2}\Big(p_i(x) + (-1)^{i-1} p_i(y)\Big).$$

It is clear that $g(x, y)$ can be written uniquely as a polynomial $P(u_1, u_2, \ldots; v_1, v_2, \ldots)$ in the u_i's and v_i's, and we want to show that no u_i appears. Now

$$g((t, x), (-t, y)) = P(u_1 + t, u_2 + t^2, \ldots; v_1, v_2, \ldots). \tag{26}$$

By considering the coefficient of t in (26) and using (25), we see that u_1 does not appear in P. Now by considering the coefficient of t^2 is follows that u_2 does not appear in P, etc. This proof is due to John Stembridge (unpublished).

Stanley Supplementary Exercise 83(e)

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83. Let $\omega_y : \Lambda(x,y) \to \Lambda(x) \otimes \Lambda(y)$ be the algebra homomorphism defined by

$$\omega_y p_n(x,y) = p_n(x) + (-1)^{n-1} p_n(y). \tag{4}$$

Equivalently, ω_y is the automorphism ω acting on the y-variables only. Write $\omega_y f(x,y) = f(x/y)$. In particular, $s_\lambda(x/y)$ is called a *super Schur function*. Let

$$\Sigma = \mathrm{im}(\omega_y) = \{f(x/y) : f \in \Lambda\},$$

a subalgebra of $\Lambda(x) \otimes \Lambda(y)$.

(e) [3–] Define a *supertableau* of shape λ to be an array T of positive integers of shape λ such that (i) the rows and columns are weakly increasing, and (ii) the diagonals from the upper left to lower right are strictly increasing (equivalently, there is no 2×2 square of equal entries). A maximal rookwise-connected subset of equal entries is called a *component* of T. Let $c(T)$ denote the number of components. For instance, if T is given by:

1	1	1	1	2	2	3
1	2	2	2	3	4	
1	2	3	3	3	4	
2	2	4	4			
3	4	4				

then T has one component of 1's, two components of 2's, three components of 3's, and two components of 4's, so $c(T) = 8$. Show that

$$s_\lambda(x/x) = \sum_T 2^{c(T)} x^T,$$

where T ranges over all supertableaux of shape λ and x^T has its usual meaning.

Original supporting material

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(a) [2–] Show that

$$s_\lambda(x/y) = \sum_{\mu \subseteq \lambda} s_\mu(x) s_{\lambda'/\mu'}(y). \tag{5}$$

83.

(a) By (7.66) we have

$$s_\lambda(x, y) = \sum_{\mu \subseteq \lambda} s_\mu(x) s_{\lambda/\mu}(y).$$

Now apply ω_y and use Theorem 7.15.6.

Original solution

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(e) From (a) it is easy to see that

$$[x^\alpha y^\beta] s_\lambda(x/y) = \langle h_\alpha e_\beta, s_\lambda \rangle.$$

This corresponds to building up the shape λ by first adding horizontal strips of sizes $\alpha_1, \alpha_2, \ldots$ and then vertical strips of sizes $\beta_1, \beta_2, \ldots$. But multiplication of symmetric functions is *commutative*, so we could first add a horizontal strip H_1 of size α_1, then a vertical strip V_1 of size β_1, then a horizontal strip H_2 of size α_2, then a vertical strip V_2 of size β_2, etc. To get $s_\lambda(x/x)$, fill in each square of $H_i \cup V_i$ with i, getting an array T, and sum all x^T's. Now the array T is precisely a supertableau. How many times does a particular supertableau T get counted? For each component C of $H_i \cup V_i$, there are exactly two ways to decompose C into a horizontal strip H followed by a vertical strip V. (The last square in the top row can be part of H or V, but there are no other choices.) Hence the number of times T gets counted is $2^{c(T)}$, completing the proof.

Stanley Supplementary Exercise 87

Richard P. Stanley

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87. [3] Let t be an indeterminate. Let $\vartheta : \Lambda \to \Lambda[t]$ be the specialization (homomorphism) defined by

$$\vartheta(p_k) = t + \sum_{i=1}^{k} \binom{k}{i} p_i.$$

Show that

$$\vartheta(s_\lambda) = \sum_{\mu \subseteq \lambda} \frac{f^{\lambda/\mu}}{|\lambda/\mu|!} \left(\prod_{u \in \lambda/\mu} (t + c(u)) \right) s_\mu,$$

where $c(u)$ denotes the content of the square u.

Original solution

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87. *Sketch of proof.* Write ϑ_n for ϑ specialized to $t = n$, where $n \in \mathbb{P}$. Note that

$$\vartheta_n(p_k)(x_1, \ldots, x_n) = p_k(x_1 + 1, \ldots, x_n + 1).$$

Hence

$$\vartheta_n(s_\lambda)(x_1, \ldots, x_n) = s_\lambda(x_1 + 1, \ldots, x_n + 1).$$

Now

$$s_\lambda(x_1 + 1, \ldots, x_n + 1) = \frac{a_{\lambda+\delta}(x_1 + 1, \ldots, x_n + 1)}{a_\delta(x_1 + 1, \ldots, x_n + 1)} = \frac{a_{\lambda+\delta}(x_1 + 1, \ldots, x_n + 1)}{a_\delta(x_1, \ldots, x_n)}.$$

By expanding the entries of $a_{\lambda+\delta}(x_1+1, \ldots, x_n+1)$ and using the multilinearity of the determinant we get (see I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, second ed., Example I.3.10 on page 47)

$$s_\lambda(x_1 + 1, \ldots, x_n + 1) = \sum_{\mu \subseteq \lambda} d_{\lambda\mu} s_\mu, \tag{27}$$

where

$$d_{\lambda\mu} = \det\left(\binom{\lambda_i + n - i}{\mu_j + n - j}\right)_{1 \le i,j \le n}.$$

We can factor out factorials from the numerators of the row entries and denominators of the column entries of the above determinant. These factorials altogether yield $\prod_{u \in \lambda/\mu}(n + c(u))$. What remains is exactly the determinant for $f^{\lambda/\mu}/|\lambda/\mu|!$ given by Corollary 7.16.3, and we obtain

$$\vartheta_n(s_\lambda)(x_1, \ldots, x_n) = \sum_{\mu \subseteq \lambda} \frac{f^{\lambda/\mu}}{|\lambda/\mu|!} \left(\prod_{u \in \lambda/\mu} (n + c(u)) \right) s_\mu(x_1, \ldots, x_n).$$

We can set $x_{i+1} = \cdots = x_n = 0$ to obtain the above equation in i variables. In other words,

$$\vartheta_n(s_\lambda)(x_1, \ldots, x_i) = \sum_{\mu \subseteq \lambda} \frac{f^{\lambda/\mu}}{|\lambda/\mu|!} \left(\prod_{u \in \lambda/\mu} (n + c(u)) \right) s_\mu(x_1, \ldots, x_i)$$

for all $n \ge i$. Both sides are polynomials in n, so they are equal as polynomials, and we can replace n with the indeterminate t. Now let $i \to \infty$.

Stanley Supplementary Exercise 110(a)

Richard P. Stanley

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110.
(a) [2+] Let $\kappa(w)$ denote the number of cycles of $w \in \mathfrak{S}_n$. Show that

$$P_n(q) := \sum_w q^{\kappa(w(1,2,\ldots,n))} = \frac{1}{n(n+1)} \left((q+n)_{n+1} - (q)_{n+1} \right).$$

where w ranges over all $(n-1)!$ n-cycles in $\mathfrak{S}_n$ and $w(1,2,\ldots,n)$ denotes the product of w with the n-cycle $(1,2,\ldots,n)$. For instance,

$$\sum_{\rho(w)=(3)} q^{\kappa(w(1,2,3))} = \frac{1}{12} \left((q+3)_4 - (q)_4 \right) = q^3 + q.$$

Hint. Use Exercise 7.70.

Original solution

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110.

(a) We may assume $q \in \mathbb{P}$. In Exercise 7.70, let $k = 3$ and $x^{(3)} = 1^q$. Take the scalar product of both sides with $p_n\left(x^{(1)}\right) p_n\left(x^{(2)}\right)$. The right-hand side becomes after some simplification

$$\mathrm{RHS} = n \sum_{\rho(w)=(n)} q^{\kappa(w(1,2,\ldots,n))}.$$

The left-hand side becomes (using Corollary 7.21.4)

$$\mathrm{LHS} = \sum_{\lambda \vdash n} \chi^{\lambda}((n))^2 \prod_{u \in \lambda} (q + c(u)).$$

Now e.g. by the Murnaghan-Nakayama rule we have

$$\chi^{\lambda}((n)) = \begin{cases} (-1)^{n-i}, & \text{if } \lambda = (i, 1^{n-i}) \\ 0, & \text{otherwise.} \end{cases}$$

From this we easily get

$$P_n(q) = \frac{1}{n} \sum_{i=1}^{n} (q + i - 1)_n = \frac{1}{n(n+1)} \left((q+n)_{n+1} - (q)_{n+1} \right).$$

Stanley Supplementary Exercise 149(a)

Richard P. Stanley

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Original statement

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(a) [3–] Let $A_1, \ldots, A_k$ be subsets of $[n]$ satisfying $\bigcup A_i = [n]$ and $A_i \cap A_j = \{1\}$ for all $i < j$. Set $a_i = \#A_i$. Show that the symmetric function

$$G_{a_1,\ldots,a_k} := \sum_{w_1 \in \mathfrak{S}_{A_1}} \cdots \sum_{w_k \in \mathfrak{S}_{A_k}} p_{\rho(w_1 \cdots w_k)} \tag{15}$$

is equal to $\prod(a_i - 1)!$ times the coefficient of $x_1^{a_1} \cdots x_k^{a_k}$ in

$$\Big(\sum_{i_1,\ldots,i_k \geq 1} p_{i_1+\cdots+i_k-k+1} x_1^{i_1} \cdots x_k^{i_k} \Big) H(x_1) \cdots H(x_k),$$

where $H(t) = \sum_{n \geq 0} h_n t^n$.

Original solution

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(a) Suppose that the cycle c of $w_1 \cdots w_k$ containing 1 intersects A_j in i_j elements, so the length of this cycle is $i_1 + \cdots + i_k - k + 1$. The cycle c will have the form

$$c = (1, u_1, u_2, \ldots, u_k),$$

where u_j is a sequence of $i_j - 1$ distinct elements of $A_j - \{1\}$. Hence there are

$$\prod_{j=1}^{k} (a_j - 1)_{i_j - 1}$$

choices for c. The elements of A_j not in c can be any permutation of size $a_j - i_j$. The generating function by cycle type of these permutations is

$$\sum_{w \in \mathfrak{S}_{a_j - i_j}} p_{\rho(w)} = (a_j - i_j)! h_{a_j - i_j}.$$

Thus the terms of the symmetric function (15) for fixed $a_1, \ldots, a_k, i_1, \ldots, i_k$ yield

$$\left(\prod_{j=1}^{k} (a_j - 1)_{i_j - 1} (a_j - i_j)! \right) \cdot p_{i_1 + \cdots + i_k - k + 1} x_1^{i_1} \cdots x_k^{i_k} h_{a_1 - i_1}(x_1) \cdots h_{a_k - i_k}(x_k).$$

Note that $(a_j - 1)_{i_j - 1}(a_j - i_j)! = (a_j - 1)!$. Divide by $\prod (a_j - 1)!$ and sum on $i_1, \ldots, i_k \geq 1$ and $a_1, \ldots, a_k \geq 0$ to complete the proof.